English • Biodiversity, adaptation and descriptive writing

Snow Leopard and Markhor

Official activities • Predator–prey balance • Hyphens and -ough • Conservation role-play • Description

Unit 10 • Complete official activities • Ecosystems + Hyphens + Conservation + Description

Text at a glance

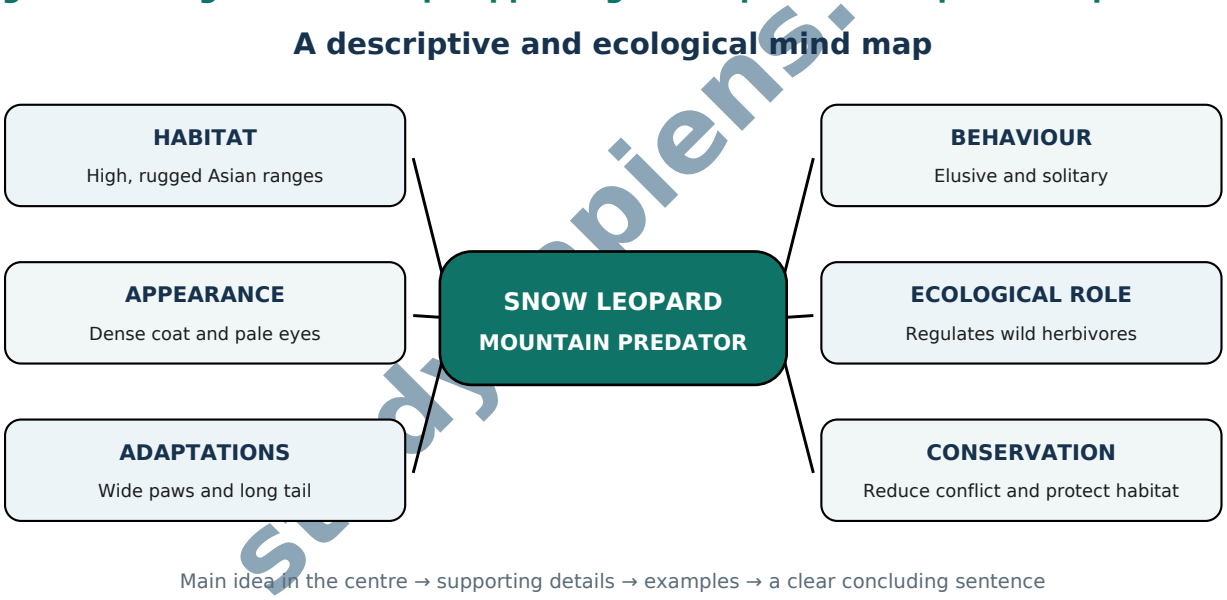
• **Snow leopard:** A shy, solitary mountain predator whose dense spotted coat, wide furred paws and long balancing tail suit steep, cold habitats.

• **Markhor:** Pakistan's national animal, a wild goat with corkscrew-shaped horns, wide climbing hooves and a seasonal coat.

• **Relationship:** Predator and prey form part of a mountain food web. Balanced numbers help maintain vegetation and wider biodiversity.

• **Threats and responses:** Poaching, habitat loss, climate change and retaliation require community incentives, habitat protection, monitoring and conflict reduction.

Figure 1 — Original mind map supporting the required descriptive composition



Main idea in the centre → supporting details → examples → a clear concluding sentence

Ecology note: The text's intended food-web point is that predators can limit excessive herbivore growth. Markhors influence vegetation through grazing and browsing, but an unchecked increase in any herbivore population can contribute to overgrazing; the animal's mere presence is not a guarantee against it.

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K-W-L: Before and After Reading

K — What I know / predict	W — What I want to know	L — What I learned
Snow leopards live in cold mountains and are difficult to see.	Which features help them move safely on snow and cliffs?	Dense fur resists cold; wide furred paws spread weight; long tails assist balance.
The markhor is Pakistan’s national animal and has spiral horns.	Where is it found, and what does it eat?	It occupies steep regions in several Pakistani mountain areas and eats grasses, shrubs and leaves.
The two animals can be predator and prey.	Why would losing either species affect the wider environment?	Food-web relationships influence herbivore numbers, vegetation and other species; losses create ripple effects.
People can harm or protect wildlife.	How can local communities benefit from conservation?	Regulated revenue, compensation, safer livestock arrangements and community participation can create incentives for protection.

While-reading margin questions. Quick answers in passage order.

Question	Answer
Why are snow leopards rarely seen by humans?	They are shy, solitary and avoid contact; their spotted coats also camouflage them among rocks.
What do markhors eat?	They are herbivores that eat grasses, shrubs and leaves.
Have conservation efforts helped snow leopard numbers?	The text reports improvement where compensation and awareness have reduced retaliatory killings. Continued monitoring is needed before treating this as a uniform trend everywhere.

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Reading and Critical Thinking

A. Answer the following questions.

1. Why is the snow leopard–markhor predator-prey relationship important?

Snow leopards obtain energy from wild prey and can prevent herbivore populations from exceeding the food and vegetation available in their habitat. Markhors, in turn, convert plant material into food for predators and influence plant growth through grazing. This interdependence supports nutrient flow and biodiversity, so a major change in either population can affect the wider mountain ecosystem.

2. How do both animals' adaptations support mountain survival?

The snow leopard's dense coat provides insulation, its wide furred paws improve movement over snow, and its long tail supports balance on ledges; camouflage and agility assist hunting. The markhor's wide hooves provide purchase on steep rock, its seasonal coat responds to temperature changes, and its strong climbing ability helps it graze and escape predators. Each adaptation matches a specific environmental challenge.

3. What are the benefits and concerns of community-led trophy hunting programmes?

• **Potential benefits:** Strictly limited permits can generate revenue for habitat work and local incomes, giving communities a financial reason to oppose poaching and maintain healthy populations.

• **Potential concerns:** Poor quotas or monitoring can harm the wrong age groups, encourage uncontrolled offtake or place income above animal welfare and ecological needs. Revenue may not reach residents fairly.

• **Required safeguards:** Independent population surveys, conservative quotas, transparent revenue sharing, anti-poaching enforcement, animal-welfare standards and regular public evaluation.

4. What might happen if either species disappeared?

Without snow leopards, wild herbivore numbers could rise in some areas, intensifying grazing pressure and changing plant communities, with further effects on soil and other animals. Without markhors, snow leopards would lose an important prey source where the species overlap and could face greater nutritional stress or livestock conflict. The precise result would depend on alternative prey, habitat condition and other predators, so the outcome is not identical everywhere.

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5. How do poaching and habitat destruction threaten the balance?

Poaching directly removes animals and can disrupt age and sex structures, while habitat loss reduces food, shelter and safe movement routes. Fewer markhors mean less wild prey; fewer snow leopards weaken predator regulation. Habitat fragmentation can also increase encounters with people and livestock, making retaliation more likely and further destabilizing the food web.

6. How is the markhor adapted to mountainous regions?

Its wide hooves help it grip steep surfaces, and its agility allows rapid movement among cliffs and ledges. A thicker winter coat provides warmth and is shed in warmer conditions. Its plant-based diet uses mountain vegetation, while the males' long twisted horns are a conspicuous species feature. These traits support feeding, temperature control and escape rather than making the animal invulnerable to human threats.

7. What more can raise awareness about endangered mountain species?

Schools can organize evidence-based wildlife projects, local-language talks and field observations without disturbing animals. Community radio, responsible documentaries and citizen science can explain food webs and conflict solutions. Tourism operators should promote distance and habitat care rather than chasing sightings. Campaigns become stronger when they show practical actions—reporting poaching, protecting livestock safely and supporting community conservation—rather than relying only on attractive photographs.

8. How did the descriptions affect your understanding of the animals and environment?

Model personal response: The descriptions made the animals seem adapted to a complete landscape rather than isolated specimens. The snow leopard's paws and tail and the markhor's hooves show how body structure reflects terrain. Their predator-prey connection also made conservation more complex: protecting one species requires prey, vegetation and local livelihoods. I now understand why community participation is part of ecological protection.

B. Choose the correct option.

Q	Answer and reason
1	(a) Herbivore populations could increase, contributing to overgrazing.
2	(b) They can generate local income and fund wildlife protection.
3	(c) Involving local communities leads to better conservation outcomes.

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Q	Answer and reason
4	(c) Balance helps keep herbivore populations in check and protects vegetation.
5	(c) Financial gain may be prioritised over animal welfare if governance is weak.

C. Complete the Problem–Cause–Effect–Solution–Analysis organiser.

Section	Relevant information
Problem	Snow leopard and markhor populations face serious threats in Pakistan’s mountains.
Cause	Poaching, habitat loss and the retaliatory killing of snow leopards by herders protecting livestock.
Effect	Ecosystem imbalance; loss of snow leopards can allow markhor numbers to rise, contributing to overgrazing and environmental damage.
Solution	Strictly regulated conservation revenue programmes and compensation for herders, alongside wider habitat and anti-poaching protection.
Analysis	The measures link wildlife survival with local incentives. Saving both species and the ecosystem requires continued monitoring, fair benefits and reduced human-wildlife conflict.

Vocabulary and Word Forms

A. Denotation and connotation in context.

Word	Denotation	Connotation in the text
elusive	difficult to find, catch or define	mystery, remoteness and the snow leopard’s ghost-like presence
heritage	valued history, traditions and natural features inherited over time	pride, identity and responsibility to preserve Pakistan’s wildlife

Word	Denotation	Connotation in the text
predator	an animal that hunts and eats other animals	power and ecological regulation, not moral wickedness
vulnerable	exposed to harm or at risk	fragility and urgency despite physical strength or impressive adaptations

A (hyphens). Add hyphens where needed.

1. The **editor-in-chief** made several notes on the manuscript.

2. The student gave a **well-written** essay.

3. The city is famous for its **state-of-the-art** infrastructure.

4. He works as a **part-time** teacher.

5. My **brother-in-law** is visiting this weekend.

B (-ough). Use pronunciation and context to complete the words.

1. The river runs **through** the city.

2. She smiled as if she hadn't gone **through** so much. (Corrected sentence; see the note below.)

3. The fabric felt very **rough** against my skin.

4. He hadn't **thought** much about the topic.

5. We don't have **enough** money to buy that car.

Correction for item 2: The printed prompt says "hadn't gone ___ so much" and labels the sound for *though*, but standard English requires **through** in "gone through so much". Use **through** for a grammatically correct answer; *though* would fit "She hadn't slept, though."

C. Make three hyphenated words with each prefix.

Prefix	Three examples
self-	self-esteem, self-reliant, self-confidence
ex-	ex-president, ex-student, ex-colleague
co-	co-author, co-founder, co-worker
non-	non-fiction, non-profit, non-violent

Some dictionaries and style guides write *co-worker*, *non-fiction* and related forms without hyphens. The hyphenated forms here satisfy the exercise; consistency matters.

D. Choose the correct -ough word.

1. though: We played outside though it was drizzling.

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2. enough: There wasn't enough sugar left for tea.

3. dough: Kiran kneaded the dough to make bread.

4. bough: The storm blew the leaves off the bough.

5. cough: He had a severe cough throughout the day.

Word	Typical British pronunciation / sound
through	/θruː/ — "oo"
though / dough	/ðəʊ/ and /dəʊ/ — "oh"
rough / enough	/rʌf/ and /ɪˈnʌf/ — "uff"
thought	/θɔːt/ — "aw"
bough	/baʊ/ — "ow"
cough	/kɒf/ — "off"

Oral Communication Skills

A. Group discussion with leader, note-taker and presenter.

Model presentation — "A Fragile Balance"

• **Main idea:** Snow leopards and markhors depend on a functioning food web, but human pressures can destabilize it.

• **Supporting details:** Snow leopards use markhors and other wild herbivores as prey; markhors use steep terrain to escape. Poaching, habitat destruction and climate change reduce both populations, while livestock losses can lead to retaliation.

• **Whole-class conclusion:** Conservation should address the entire relationship—prey, predators, habitat and local livelihoods—rather than focusing on one animal in isolation.

B. Research and tell a short conservation story.

Model research story — Reducing conflict with WWF-Pakistan

• A herder in snow leopard habitat may first see the cat as a threat to family income. WWF-Pakistan's conservation work addresses that practical conflict rather than simply asking communities to tolerate losses. Its official overview describes support for predator-proof livestock pens and compensation schemes for verified losses. These measures can reduce attacks and soften the pressure for retaliation, while education explains the predator's ecological role. The work matters because coexistence protects wildlife and livelihoods together; success still requires fair administration and continued local participation.

Research reference: WWF-Pakistan, "Snow Leopard", official conservation overview: wwfpak.org/our_work_/wildlife_2/snow_leopard_/ (accessed 7 November 2026). For a live class presentation, add

one locally relevant example and evaluate how the programme is monitored.

C. Role-play a community conservation meeting.

Model dialogue

- **Wildlife conservationist:** “We need connected habitat, reliable population monitoring and enough wild prey. Removing either predator or prey will alter the ecosystem.”
- **Community member:** “Our livestock losses are real. Conservation will work only if compensation is prompt, corrals are safer and residents share decisions.”
- **Government official:** “We can support anti-poaching patrols, transparent compensation and independent review of any regulated hunting programme.”
- **Local entrepreneur:** “Responsible eco-tourism and local crafts can create income, but visitor numbers and waste must be controlled so business does not damage the habitat.”
- **Wildlife conservationist:** “Then our shared plan is habitat protection, conflict prevention, fair benefits and published results.”

D. Discuss challenges and debate whether predator and prey can both be saved.

Community involvement. How does it help, and which challenges need attention?

Residents possess local knowledge and can monitor remote areas more consistently than occasional outside teams. Fair incentives and participation build trust. Challenges include unequal benefit sharing, livestock losses, weak enforcement and disagreement over hunting. Transparent records, community representation and independent ecological monitoring can address these concerns.

Predator side. Why prioritize the snow leopard?

As a top predator, the snow leopard influences wild herbivore numbers and vegetation across a broad landscape. Its rarity and conflict with livestock make targeted protection urgent. Predator-proofing and compensation therefore deliver ecosystem and community benefits.

Prey side. Why is the markhor equally important?

The markhor supplies prey, influences plant communities and carries national cultural significance. Its recovery can support local pride and conservation livelihoods. However, its value depends on suitable habitat and balanced numbers rather than unlimited population growth.

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Synthesis. Can both be saved?

Yes, and they should be managed together. Protecting habitat, wild prey and predators while reducing livestock conflict addresses their interdependence. A programme that saves one by degrading the other would fail the unit's central ecological lesson.

E. Attentive listening and standard pronunciation.

Word	Pronunciation guide
elusive	/ɪˈluːsɪv/
ecosystem	/ˈiːkəʊˌsɪstəm/
herbivore	/ˈhɜːbɪvɔː(r)/
vulnerable	/ˈvʌlnərəbəl/
conservation	/ˌkɒnsəˈveɪʃən/
altitude	/ˈæltɪtjuːd/

Stress the bold-sounding syllable in speech, pause after commas and use a falling tone for complete statements. Listen for meaning before attempting to imitate pronunciation.

Writing Skills: Descriptive Composition

Steps 1–2. Brainstorm and mind-map habitat, appearance, adaptations, behaviour and importance.

Branch	Specific snow leopard notes
Habitat	High mountains, cold temperatures, rocky ledges, snow, remote terrain
Appearance	Dense pale spotted coat, muscular body, wide paws, very long thick tail
Adaptations	Insulation, camouflage, snow movement, balance, agile hunting
Behaviour	Elusive, solitary, avoids people, hunts wild herbivores
Importance	Top predator, food-web balance, natural heritage, conservation indicator

Step 3. Write the first draft from general to specific.

Model descriptive composition — The Snow Leopard

• High in the rugged mountain ranges of Pakistan lives the snow leopard, a powerful yet rarely seen predator. It inhabits cold, rocky landscapes where survival requires insulation, balance and patience. As part of the mountain food web, it is significant not only for its beauty but also for its influence on wild herbivore populations.

• The animal's most noticeable feature is its thick, pale coat patterned with dark rosettes and spots. This dense fur protects it from severe cold, while the pattern breaks up its outline among stones and shadows. Its muscular body suggests strength, and its pale eyes give its face a watchful expression. A remarkably long, thick tail and broad, fur-covered paws make the animal immediately distinctive.

• Every major physical trait serves a purpose in this habitat. The wide paws spread the leopard's weight and help it move across snow, while the tail assists balance on narrow ledges and steep slopes. Camouflage allows it to approach prey without easy detection. Agility and careful movement are as important as force in a landscape where a misplaced step can be costly.

• Snow leopards are generally shy and solitary, avoiding human contact whenever possible. They hunt markhors where ranges overlap, as well as other wild prey such as ibex and wild sheep. By influencing herbivore numbers, predators can help protect vegetation from excessive grazing pressure. The cat is therefore part of an ecological relationship rather than merely a frightening hunter.

• Poaching, habitat disturbance and conflict over livestock threaten this delicate balance. Protecting the snow leopard requires connected habitat, wild prey and practical support for herders. Predator-proof enclosures, fair compensation and community monitoring can reduce retaliation. The snow leopard's survival would preserve not only a magnificent animal but also the health and character of Pakistan's mountain wilderness.

Original Exam-Practice MCQs

1. Which adaptation helps a snow leopard balance on ledges?

(a) long tail (b) wide hooves
(c) spiral horns (d) seasonal antlers

Answer: (a) long tail — The long tail acts as a balancing aid.

2. The markhor is:

(a) a carnivore (b) a herbivore
(c) an amphibian (d) a reptile

Answer: (b) a herbivore — It eats grasses, shrubs and leaves.

3. Which is a direct human threat to both species?

(a) poaching (b) photosynthesis
(c) hibernation (d) pollination

Answer: (a) poaching — Illegal killing reduces populations.

4. Why can losing snow leopards affect vegetation?

(a) they eat all plants (b) herbivore regulation may weaken
(c) they produce oxygen (d) they build nests for goats

Answer: (b) herbivore regulation may weaken — Reduced predation can increase grazing pressure.

5. A key concern with trophy programmes is:

(a) automatic habitat creation (b) financial priorities may override welfare
(c) they eliminate all poaching by definition (d) they require no monitoring

Answer: (b) financial priorities may override welfare — Strong governance and safeguards are essential.

6. “Elusive” most nearly means:

(a) easy to find (b) difficult to find
(c) dangerous to touch (d) noisy

Answer: (b) difficult to find — It suits the rarely seen snow leopard.

7. Which word is correctly hyphenated?

(a) editor in chief (b) well-written
(c) state of the art (d) brother in law

Answer: (b) well-written — The compound adjective before “essay” takes a hyphen.

8. Which -ough word has the /ʌf/ sound?

(a) through (b) though
(c) rough (d) dough

Answer: (c) rough — Rough and enough share the “uff” sound.

9. “Dough” is used for:

(a) bread mixture (b) a tree branch
(c) a severe cough (d) light rain

Answer: (a) bread mixture — Kneading dough prepares bread.

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10. The best conservation conclusion is:

(a) protect predator, prey, habitat and livelihoods together (b) remove all predators

(c) focus only on attractive animals (d) ignore local residents

Answer: (a) protect predator, prey, habitat and livelihoods together — Interdependence requires integrated management.

Points to Remember

Snow leopard: elusive predator; dense coat, wide furred paws and long tail suit cold, steep terrain.

Markhor: national animal and herbivore; spiral horns, wide hooves and seasonal coat.

Food web: predators can limit excessive herbivore growth; prey, vegetation and habitat remain interconnected.

Threats: poaching, habitat destruction, climate change and livestock-related retaliation.

Conservation: community incentives, fair compensation, habitat protection, monitoring and transparent governance.

Hyphens: editor-in-chief, well-written, state-of-the-art, part-time, brother-in-law.

-ough: learn each word's sound individually; the spelling string has several pronunciations.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	a	2	b	3	a	4	b	5	b
6	b	7	b	8	c	9	a	10	a

How to attempt this unit in the paper

- For adaptation questions, link each physical feature to a survival function.
- For ecosystem answers, use "can" or "may" rather than claiming that every local outcome is certain.
- For trophy hunting, present benefits and safeguards as well as ethical and governance concerns.
- Use **through** in the corrected phrase "gone through so much"; the printed prompt is internally inconsistent.
- In description, move from habitat and general identity to appearance, adaptations, behaviour and significance.

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